MIL-D-0050082K(EA)
8 August 1983
SUPERSEDING
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7 October 1981
USED IN LIEU OF
MIL-D-50082D
30 September 1964

MILITARY SPECIFICATION

# DISKS, INLET AND NOSECUP VALVE

This limited coordination military specification has been prepared by the US Army Armament, Munitions and Chemical Command based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-D-50082D. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in acquisition.

## 1. SCOPE

1.1 Scope. This specification covers silicone rubber valve disks.

## 2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

ZZ-R-765 - Rubber, Silicone.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4240

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STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

| | | |
|---|---|---|
| B5-2-1034 | - | Disk, Valve, Nonmetallic, Nosecup and Inlet. |
| C5-2-1016 | - | Cover, Cap-Inlet Valve. |
| B136-14-170-3 | - | Air Leakage Tester, Flow Type, Q211, Assembly. |
| E136-19-22 | - | Flexibility Tester, Natural Rubber, Q37, Assembly. |
| B136-44-1-2 | - | Resistance Indicator, Inlet and Outlet Valve, Q212, Assembly. |

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. The silicone rubber used in the manufacture of the disks shall conform to ZZ-R-765, class 3b, grade 30 except the flexure resistance requirement and test are deleted.

3.2 Manufacture. The disks shall be die cut from molded sheet stock. The disks shall be manufactured in accordance with the Drawing B5-2-1034.



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3.3 Test slabs. Six slabs 6 by 6 by 0.075 ± 0.005 inches and one test slab 6 by 6 by 0.50 inches shall be made from the preproduction sample stock and from the stock used for each regular production lot. The slabs shall be identical in composition and cured to the same state of the disk.

3.4 Flexibility. The valve disks shall have a flexibility of no less than 8 units at 24° ± 3°C and shall retain 85 percent (min) of that flexibility at -45° ± 3°C when tested as specified in 4.4.4.1.2.

3.5 Distortion. The valve disk shall be free from distortion when tested as specified in 4.4.4.2.

3.6 Airflow resistance. When assembled to the valve seat, the disks shall have a resistance not to exceed a pressure differential of 5 millimeters of water at an airflow rate of 42.5 liters per minute when tested as specified in 4.4.4.3.

3.7 Leakage. When assembled to the valve seat, the disks shall not leak more than 150 milliliters of air per minute at a pressure differential of one inch of water when tested as specified in 4.4.4.4.

3.8 Workmanship. The disks shall have a smooth finish on both sides; shall have clean cut peripheral and hole edges; and shall be free from tackiness, imperfections, such as porous areas, cracks, checks, abrasions, and blisters, foreign matter (grease, oil, dirt, or ash), and damage, such as tears, slits, and punctures.

3.9 Preproduction. Prior to the start of regular production and following any change in regular production methods or rubber formulation, a preproduction sample of disks and test slabs shall be produced in accordance with this specification for examination and test.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 100 disks and test slabs (see 3.3) shall be produced using the same methods, materials, equipment and essentially the same manufacturing conditions as will be used during regular production.

4.3.2 Inspection procedure.

4.3.2.1 For examination. The preproduction sample disks shall be inspected and tested for compliance with the drawings and requirements of this specification.

4.3.2.2 For destructive tests. The sample test slabs shall be tested in accordance with table V of ZZ-R-765.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examinations and tests specified in 4.3.2.1 and 4.3.2.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the disks produced by one manufacturer from one batch (see 6.3) of raw material and manufactured under essentially the same manufacturing conditions.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. The sample test slabs shall be prepared as specified in 3.3.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample disks shall be examined and tested in accordance with the classification of defects and with MIL-STD-105.



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4.4.3.2 For destructive tests. The sample test slabs shall be tested in accordance with table V of ZZ-R-765. Acceptance number is zero.

4.4.3.3 Classification of defects.

(a) Disk, valve, nonmetallic, nosecup and inlet (Drawing B5-2-1034).

| Categories | | Defects | Acceptance standards |
|---|---|---|---|
| Critical: | | None defined | |
| Major: | AQL 0.65 percent defective | | |
| 101 | | Airflow resistance | 4.4.4.3 |
| *102 | | Leakage | 4.4.4.4 |
| | AQL 2.5 percent defective | | |
| 103 | | Distortion | 4.4.4.2 |
| 104 | | Thickness incorrect | |
| 105 | | Center hole location incorrect | |
| 106 | | Center hole diameter incorrect | |
| 107 | | Disk diameter incorrect | |
| 108 | | Tackiness | |
| 109 | | Workmanship (see 3.8) | |
| Minor: | AQL 4.0 percent defective | | |
| 201 | | Color incorrect | |

*The leakage test shall be performed in compliance with this specification unless the contract (see 6.2) requires the contractor to perform the leakage test at a higher stage of assembly.

4.4.4 Tests.

4.4.4.1 Physical properties.

4.4.4.1.1 Preparation of test specimens. The specimens shall be prepared from test slabs (see 3.3) for testing in accordance with table V of ZZ-R-765.

4.4.4.1.2 Flexibility. Flexibility shall be determined with Q37 Natural Rubber Flexibility Tester (Dwg E136-19-22). Flexibility shall be read on the grid of the apparatus. Four specimens shall be tested at 24 $\pm$ 2°C and then at -45° $\pm$ 3°C after being conditioned for 24 hours at that temperature.



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4.4.4.2 Distortion. Evenly and thoroughly spray water onto a blotter being careful not to saturate the blotter to the point that standing water is observed. If blotter is too wet, the disks will adhere to the blotter. Gently place the dry disks onto the blotter and observe for distortion of the disk. Distortion at any time within a ten minute period of any of the disks would be cause for rejection. The disks shall not be reversed on the wet blotter during the test period.

4.4.4.3 Air flow resistance. The valve disks shall be tested for airflow resistance with the Q212 Indicator, Inlet and Outlet Valve Resistance, Assembly (Drawing B136-44-1-2) after it is assembled to Cover, Cap-Inlet Valve (Drawing C5-2-1016).

4.4.4.4 Leakage. The valve disks shall be tested for leakage with the Q211 Tester, Flow Type, Air Leakage, Assembly (Drawing B136-14-170-3) after it is assembled to Cover, Cap-Inlet Valve (Drawing C5-2-1016).

5. PACKAGING

5.1 Unit packing, packing and marking, interplant shipment (see 6.6). The disks shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed. Std. No. 123 and shall include the contractor's name, address, and contract number.

5.2 Repair parts. When this item is being procured for storage and issue as a repair part, preservation, unit packing, packing and marking shall be as specified on the packaging data sheet which is identified by its National Stock Number.

6. NOTES

6.1 Intended use. The disks are intended for universal use in the nosecup valve and inlet valve of chemical-biological masks. They are also to be used in place of the winterization disks.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicability of leakage test (see 4.4.3.3).
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not to proceed with production.



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6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process and subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Valve seats. Valve seats required for conducting airflow resistance and leakage tests will be furnished by the Government.

6.5 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 216 and 234. When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.6 Interplant shipments. Packaging for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments and shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A792



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions -- Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-D-0050082K(EA) | DISKS, INLET AND NOSECUP VALVE |

**3a. NAME OF SUBMITTING ORGANIZATION**

**b. ADDRESS** *(Street, City, State, ZIP Code)*

**4. TYPE OF ORGANIZATION** *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

**5. PROBLEM AREAS**

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

**6. REMARKS**

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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